



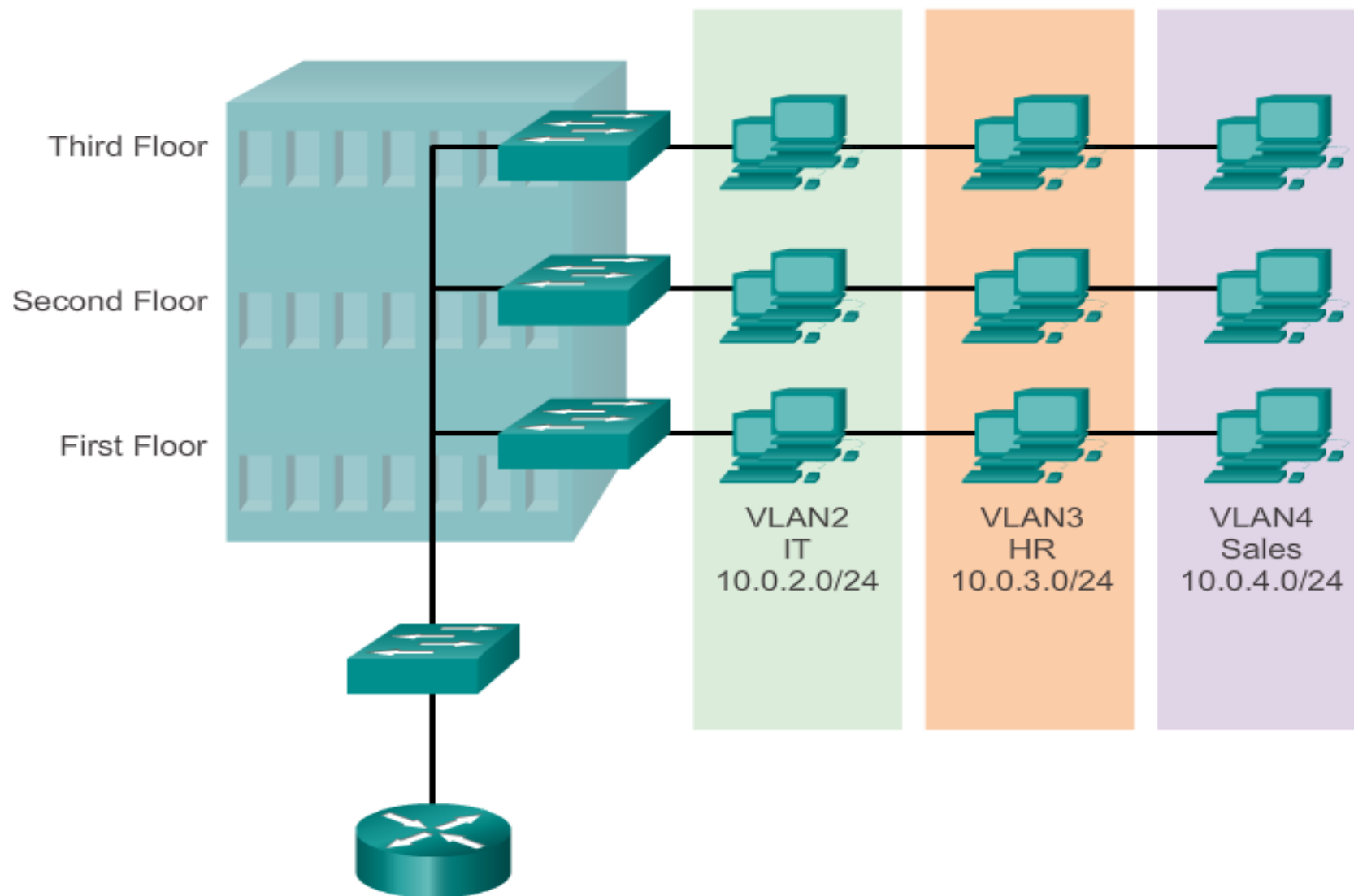
Chapter Four

Switch Configuration



VLAN

- VLAN (virtual LAN) is a **logical partition** of a network
- Each VLAN is a **broadcast domain**, usually with its own IP network.
- VLANs are mutually **isolated** and packets can only pass between them through a router
- The partitioning of the layer 2 network takes inside a layer 2 device, **usually a switch**.
- Hosts on the same VLAN are **unaware of the VLAN's existence**.





- In a **switched network**, a device can be assigned to a VLAN based on its, **MAC address**, **IP address**, or the **applications** that the device most frequently uses.
- Administrators **assign membership** in a VLAN either **statically** or **dynamically**.
- **Static VLAN membership** requires an administrator to manually assign each switch port to a specific VLAN.

Example, port fa0/3 may be assigned to VLAN 20.

- Any device that plugs into port **fa0/3** automatically becomes a member of **VLAN 20**.
- This type of VLAN membership is the **easiest** to configure and is also the **most popular**, however, it **requires** the most administrative support for adds, moves and changes.



- **Dynamic VLAN membership** requires a VLAN management policy server (VMPS).
- The VMPS contains a database that **maps MAC addresses** to VLAN assignments.
- **When a device plugs** into a switch port, the VMPS searches the database for a match of the MAC address and **temporarily** assigns that **port** to the appropriate VLAN.
- Dynamic VLAN membership requires **more configuration** but creates a structure with much **more flexibility** than static VLAN membership.
- In dynamic VLAN, **moves, adds, and changes** are **automated** and do not require intervention from the administrator.



Configuring VLAN

- Whether VLANs are created statically or dynamically, the **maximum number** of VLANs depends on the type of switch and the IOS. By default, **VLAN1** is the management VLAN.

Management VLAN

- A management VLAN is any VLAN configured to access the **management** capabilities of a switch.
- **VLAN 1** is the management VLAN by default.
- To **create the management VLAN**, the switch virtual interface (SVI) of that VLAN is assigned an **IP address** and **subnet mask**, allowing the switch to be managed via **HTTP, Telnet, SSH, or SNMP**
- An administrator will **use the IP address of the management VLAN** to configure the switch remotely.



- When a VLAN is created, it is assigned a **number** and a **name**.
- The VLAN number is **any number** from the range available on the switch, except for **VLAN1**.
- Set the following commands to create a VLAN using global configuration mode:

Switch(config)#vlan vlan_number

Switch(config-vlan)#name vlan_name

Switch(config-vlan)#exit



- **Assign ports** to be members of the VLAN.
- By default, **all ports** are initially **members of VLAN1**.
- Assign ports one at a time or as a range.

- **Assign port to VLAN one by one:**

Switch(config)#interface **fa#/#**

Switch(config-if)#switchport access vlan vlan_number

Switch(config-if)# exit

- **Assign by range**

Switch(config)#interface range fa#/start_of_range - end_of_range

Switch(config-if)#switchport access vlan vlan_number

Switch(config-if)#exit



Verifying vlan configuration

- To **verify**, maintain, and troubleshoot VLANs, it is important to understand the key show commands that are available in the Cisco IOS.
- The following commands are used to verify and maintain VLANs:
show vlan brief
- Displays a detailed **list of all of the VLAN** numbers and names currently **active on the switch**, along with the ports associated with each one

```
Switch#show vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/4, Fa0/5 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
27	accounting	active	Fa0/13
28	engineering	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1		1000001	1500							



■ show vlan id id_number

```
Switch#show vlan id 28
```

VLAN	Name	Status	Ports
28	engineering	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Tran1	Trans2
28	enet	100028	1500	-	-	-	-	-	0	0

RemoteSPAN VLANS

Disabled

- Displays information pertaining to a specific VLAN, based on ID number

■ Show vlan name vlan_name

```
Switch#show vlan name engineering
```

VLAN	Name	Status	Ports
28	engineering	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9

VLAN	Type	SAID	MTU	Parent	Ring	NoBridgeNo	Stp	BrdgMode	Tran1
28	enet	100028	1500	-	-	-	-	-	0

Remote SPAN VLANS

Disabled

me



■ To delete a VLAN:

Switch(config)#no vlan vlan_number

To disassociate a port from a specific VLAN:

Switch(config)#interface fa##/##

Switch(config-if)#no switchport access vlan vlan_number

```
Switch(config)#interface fa0/8
Switch(config-if)#no switchport access vlan 28
Switch(config-if)#exit
Switch(config)#no vlan 27
Switch(config)#end
Switch#show vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/4, Fa0/5, Fa0/8 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24

I am deleting VLAN 27. I am also disassociating port 8 from VLAN 28.





Trunk port

- A VLAN has three major functions:
 - ✓ Limits the size of broadcast domains
 - ✓ Improves network performance
 - ✓ Provides a level of security

- To take full advantage of the benefits of VLANs, they are extended across multiple switches.
- Switch ports can be configured for two different roles.
 - ✓ **Access port**

 - ✓ **A trunk port.**

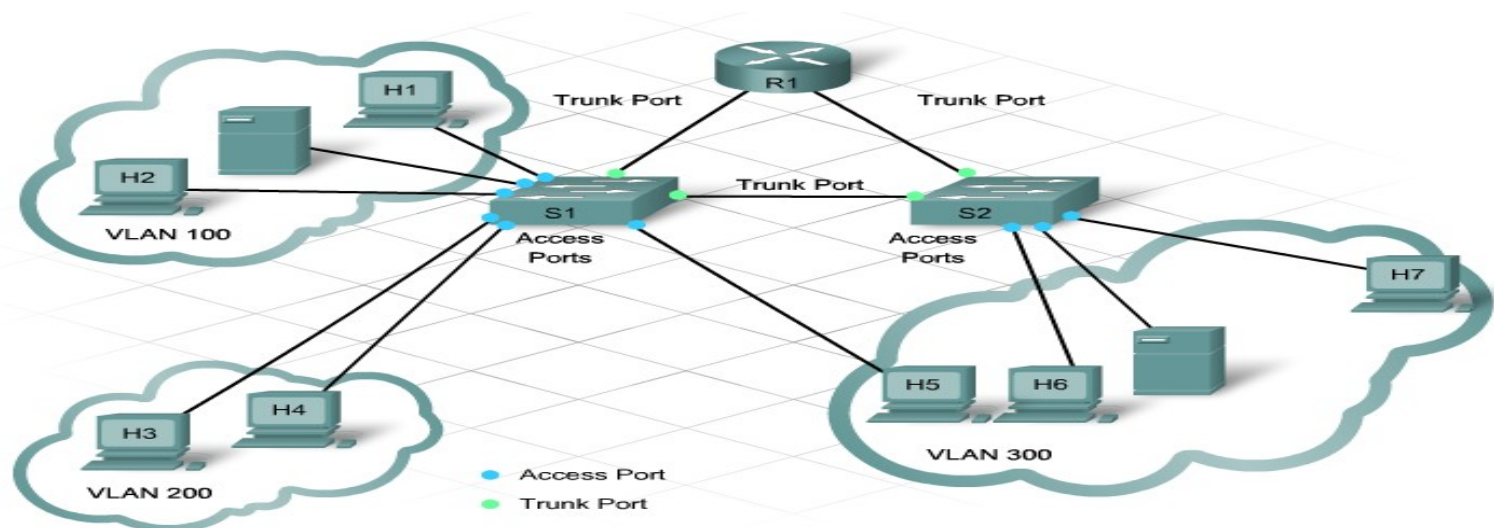


□ Access Port

- An access port belongs to **only one VLAN**.
- Typically, **single devices** such as PCs or servers connect to this type of port.
- Access ports also carry traffic that comes from **only the VLAN** assigned to the port.

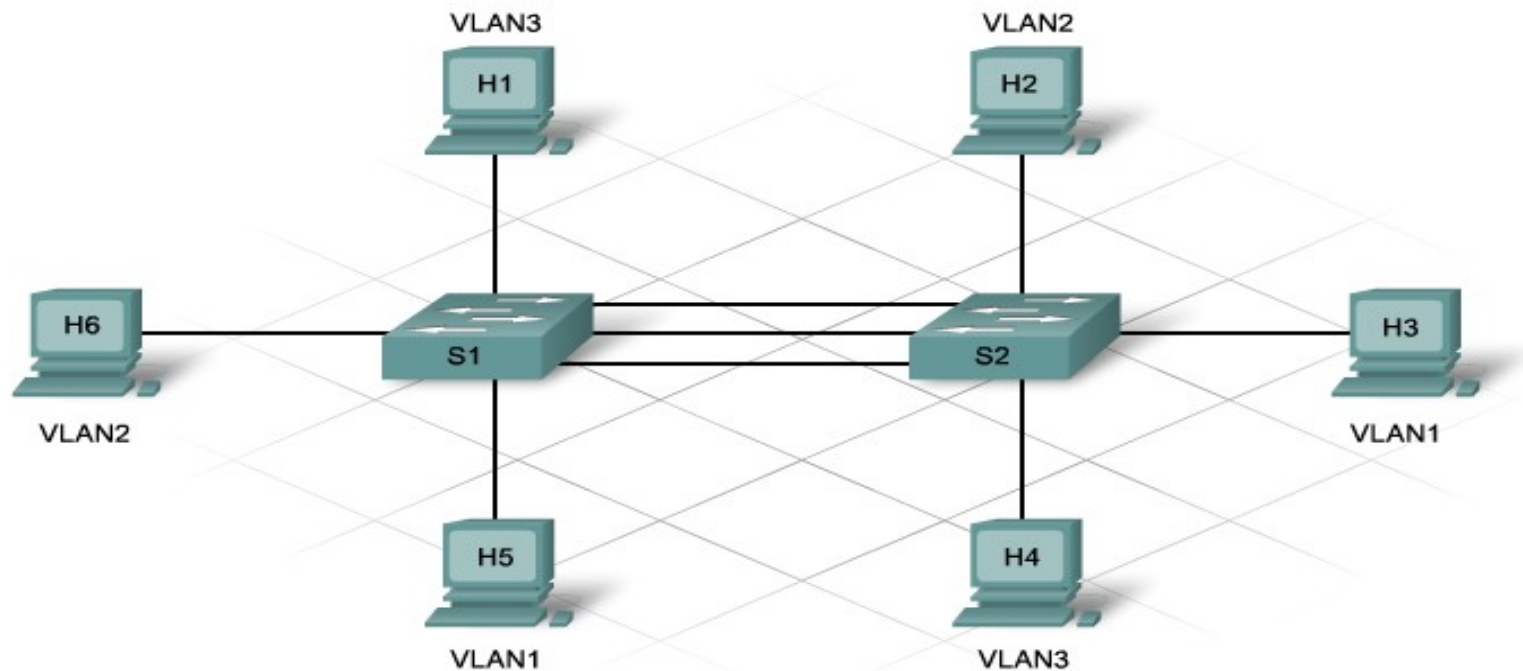
Trunk Port

- A trunk port is a **point-to-point** link between the **switch** and another networking device.
- Trunks carry the traffic of **multiple VLANs** over a single link and allow VLANs to reach across an entire network.
- Trunk ports are necessary to **carry the traffic from multiple** VLANs between devices when connecting either two switches together, a switch to a router.





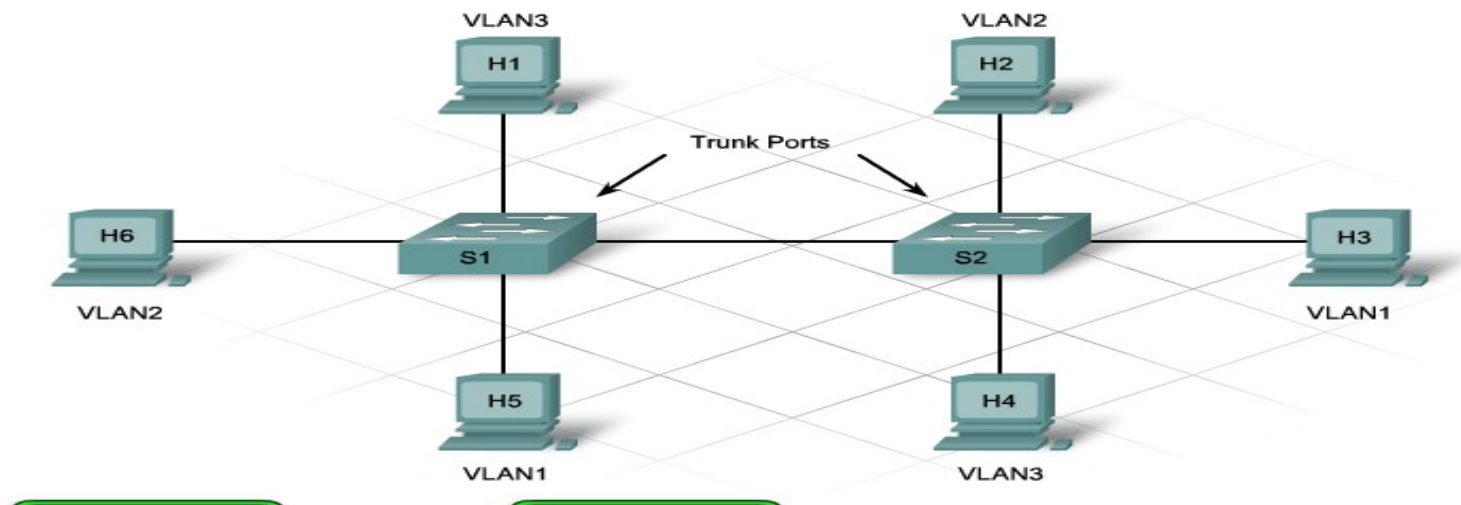
- **Without trunk** ports, each VLAN requires a **separate connection** between switches.
- For example, an enterprise with 100 VLANs requires 100 connecting links.



- This type of arrangement does **not scale** well and is **very expensive**.



- **Trunk links** provide a solution to this problem by **transporting** traffic from **multiple VLANs** on the same link.
- **When multiple** VLANs travel on the same link, they **need VLAN identification**.
- A **trunk port** supports frame **tagging**. Frame tagging adds VLAN information to the frame.
- Cisco developed a proprietary **frame tagging protocol** called **Inter-Switch Link (ISL)**.





- Switch ports are **access ports** by default.
- To configure a switch port as a **trunk port**, use the following commands:

Switch(config)#interface fa ###

Switch(config-if)#switchport mode trunk

Switch(config-if)#switchport trunk encapsulation dot1q

```
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk encapsulation dot1q
```





- Switches have the capability to **detect the type of link configured at the other end.**
- **Based on the attached device**, the link configures itself as either a **trunk** port or an **access** port.

Switch(config-if)#switchport mode dynamic {desirable | auto}

- **switchport mode dynamic auto:** The interface becomes a **trunk interface** if the neighboring interface is set to **trunk or desirable mode**.
- **switchport mode dynamic desirable:** The interface becomes a **trunk interface** if the neighboring interface is set to **trunk, desirable, or auto mode**.



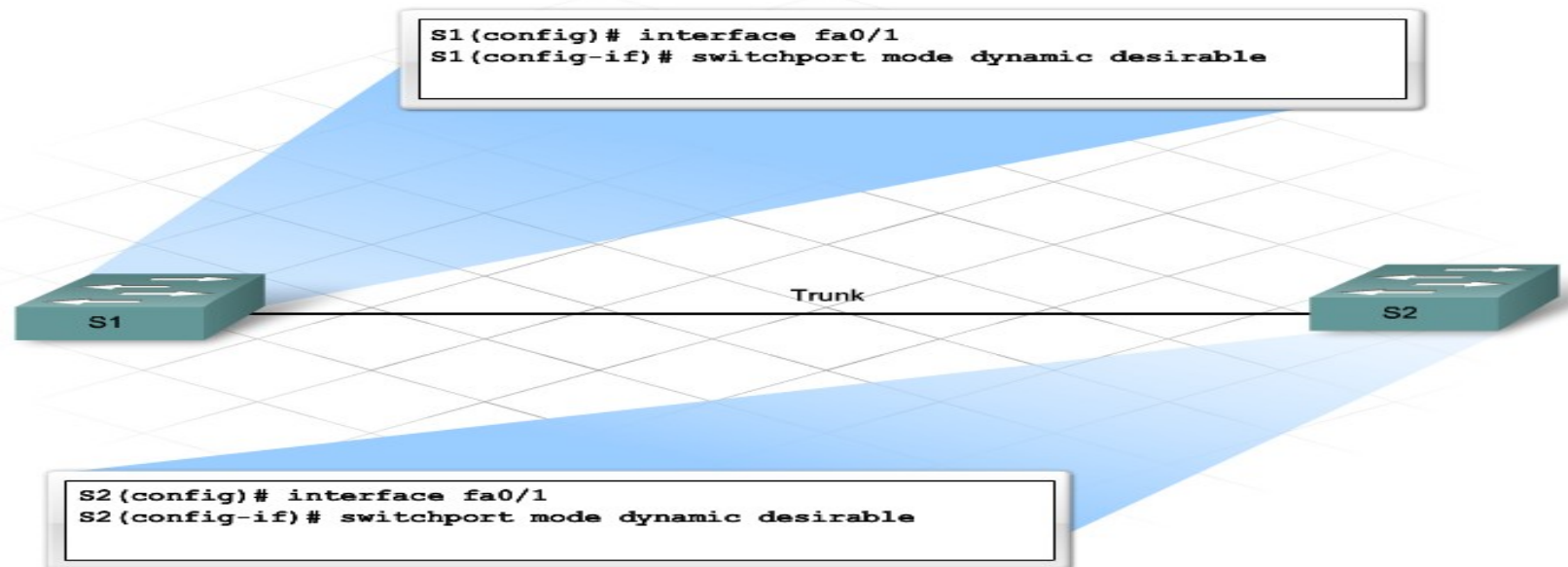
To return a trunk port to an access port, issue either of the following commands:

Switch(config)#interface fa#/#

Switch(config-if)#no switchport mode trunk or

Switch(config-if)#switchport mode access

Switch(config-if)#switchport mode access





❑ Native VLAN

- To accommodate untagged traffic, a special VLAN called a **native VLAN is available**.
- **Untagged frames** received on the 802.1Q trunk port will become members of the **native VLAN**.
- On Cisco Catalyst switches, VLAN 1 is the **native VLAN** by default.
- VLAN is identified by a “**tag**” in the frame.
- Native VLAN **does not have a tag**.
- On an 802.1Q trunk, use the following command to assign the native VLAN ID on a physical interface:

Switch(config-if)#dot1q native vlan vlan-id



Switch Security

❑ Secure Remote Access (SSH)

- Telnet uses unsecure plaintext of the username and password as well as the data transmitted.
- SSH is more secure because it provides an encrypted management connection

```
S1# configure terminal
S1(config)# ip domain-name cisco.com
S1(config)# crypto key generate rsa
The name for the keys will be: S1.cisco.com
...
How many bits in the modulus [512]: 1024
...
S1(config)# username admin secret ccna
S1(config-line)# line vty 0 15
S1(config-line)# transport input ssh
S1(config-line)# login local
S1(config-line)# exit
S1(config)# ip ssh version 2
S1(config)# exit
S1#
```



Switch security

❑ Secure Unused Ports

The `interface range` command can be used to apply a configuration to several switch ports at one time.

Disable Unused Ports



```
S1# show run
Building configuration...
...
version 15.0
hostname S1
...
interface FastEthernet0/4
 shutdown
!
interface FastEthernet0/5
 shutdown
!
interface FastEthernet0/6
 description web server
!
interface FastEthernet0/7
 shutdown
!
...
```

Disable unused ports using the **shutdown** command.



Switch security

□ Port Security

- Port security **limits** the number of valid MAC addresses allowed to transmit data through **a switch port**.
- If a port has port security enabled and **an unknown MAC address sends data**, the switch **presents a security violation**.
- Default number of secure MAC addresses **allowed is 1**.
- Methods use to configure MAC addresses within port security:
- **Static secure MAC addresses** – manually configure
`switchport port-security mac-address mac-address`



Switch security

- **Dynamic secure MAC addresses** – dynamically learned, stored in its MAC address table and removed if the **switch restarts**
- **Sticky secure MAC addresses** – dynamically learned and added to the **running configuration** (which can later be saved to the **startup-config** to permanently retain the MAC addresses)

switchport port-security mac-address sticky *mac-address*

- **Note:** Disabling sticky learning converts **sticky MAC addresses** to **dynamic secure addresses** and removes them from the running-config.



Switch Port Security

■ Port Security: Violation Modes

- ❑ **Protect** – data from unknown source MAC addresses are dropped; a security notification **IS NOT** presented by the switch
- ❑ **Restrict** - data from unknown source MAC addresses are dropped; a security notification **IS** presented by the switch and the **violation counter increments**.
- ❑ **Shutdown** – (default mode) interface becomes error-disabled and port LED turns off. The **violation counter increments**.

Violation Mode	Forwards Traffic	Sends Syslog Message	Displays Error Message	Increases Violation Counter	Shuts Down Port
Protect	No	No	No	No	No
Restrict	No	Yes	No	Yes	No
Shutdown	No	No	No	Yes	Yes

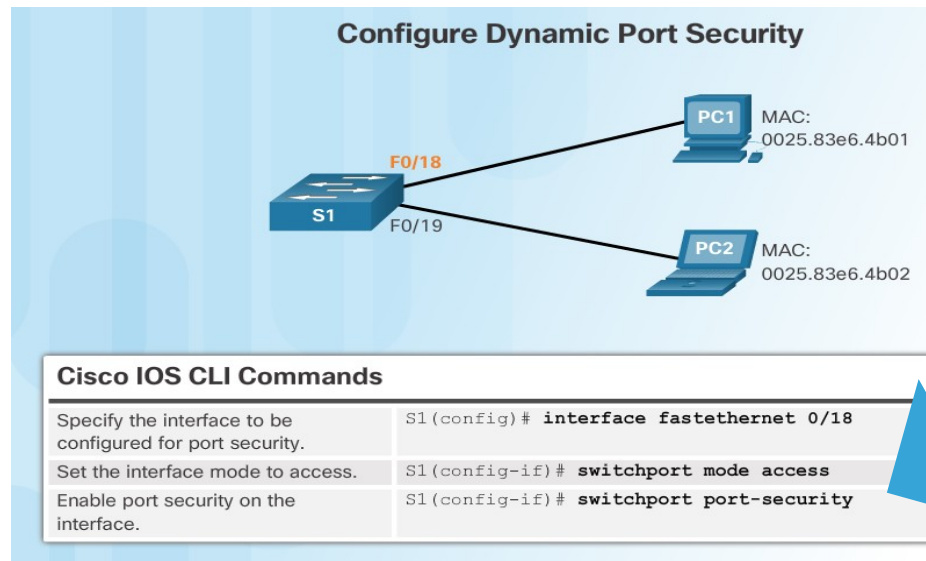


Switch Security

■ Port Security: Configuring

Feature	Default Setting
Port security	Disabled on a port
Maximum number of secure MAC addresses	1
Violation mode	Shutdown. The port shuts down when the maximum number of secure MAC addresses is exceeded.
Sticky address learning	Disabled

- Before configuring port-security features, place the port in access mode and use the **switchport port-security** interface configuration command to enable port security on an interface



Most common configuration error is to forget this command!



Port security

- Switch(config-if)#switchport port-security max ?

<1-132> Maximum addresses

Switch(config-if)#switchport port-security mac-address ?

H.H.H 48 bit mac address

sticky Configure dynamic secure addresses as sticky

- Switch(config-if)#switchport port-security violation ?

protect Security violation protect mode

restrict Security violation restrict mode

shutdown Security violation shutdown mode

- Switch(config-if)#switchport port-security maximum 10



Port Security: Configuring (Cont.)

Configure Sticky Port Security



Cisco IOS CLI Commands

Specify the interface to be configured for port security.	<code>S1(config)# interface fastethernet 0/19</code>
Set the interface mode to access.	<code>S1(config-if)# switchport mode access</code>
Enable port security on the interface.	<code>S1(config-if)# switchport port-security</code>
Set the maximum number of secure addresses allowed on the port.	<code>S1(config-if)# switchport port-security maximum 10</code>
Enable sticky learning.	<code>S1(config-if)# switchport port-security mac-address sticky</code>

Most common configuration error is to forget this command!



Port Security: Verifying

- Use the **show port-security interface** command to verify the maximum number of MAC addresses allowed on a particular port and **how many of those addresses were learned dynamically** using sticky.

```
S1# show port-security interface fastethernet 0/18
```

```
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses    : 1
Total MAC Addresses      : 1
Configured MAC Addresses : 0
Sticky MAC Addresses     : 0
Last Source Address:Vlan : 0025.83e6.4b01:1
Security Violation Count : 0
```

```
S1# show port-security interface fastethernet 0/19
```

```
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses    : 10
Total MAC Addresses      : 1
Configured MAC Addresses : 0
Sticky MAC Addresses     : 1
Last Source Address:Vlan : 0025.83e6.4b02:1
Security Violation Count : 0
```



Port Security: Verifying (Cont.)

- Use the **show running-config** command to see **learned MAC addresses** added to the configuration

```
S1# show run | begin FastEthernet 0/19
interface FastEthernet0/19
  switchport mode access
  switchport port-security maximum 10
  switchport port-security
  switchport port-security mac-address sticky
  switchport port-security mac-address sticky 0025.83e6.4b02
```

- The **show port-security address** command shows how MAC addresses were learned on a particular port.

```
S1# show port-security address
Secure Mac Address Table
-----
Vlan    Mac Address      Type             Ports    Remaining Age
(mins)
----    -
1       0025.83e6.4b01   SecureDynamic    Fa0/18   -
1       0025.83e6.4b02   SecureSticky     Fa0/19   -
-----
```



Ports in Error Disabled State

- **Switch console messages display when a port security violation occurs.** Notice the port link status changes to down.

```
Sep 20 06:44:54.966: %PM-4-ERR_DISABLE: psecure-violation error detected on Fa0/18,
putting Fa0/18 in err-disable state
Sep 20 06:44:54.966: %PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred,
caused by MAC address 000c.292b.4c75 on port FastEthernet0/18.
Sep 20 06:44:55.973: %LINEPROTO-5-PPDOWN: Line protocol on Interface
FastEthernet0/18, changed state to down
Sep 20 06:44:56.971: %LINK-3-UPDOWN: Interface FastEthernet0/18, changed state to down
```




Ports in Error Disabled State (Cont.)

- Check the port status and the port security settings.

```
S1# show interface fa0/18 status
Port Name      Status      Vlan  Duplex  Speed  Type
Fa0/18         err-disabled 1      auto    auto    10/100BaseTX

S1# show port-security interface fastethernet 0/18
Port Security           : Enabled
Port Status             : Secure-shutdown
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 0
Configured MAC Addresses : 0
Sticky MAC Addresses    : 0
Last Source Address:Vlan : 000c.292b.4c75:1
Security Violation Count : 1
```

- Do not re-enable a port until the security threat is investigated and eliminated.
- Notice that you must first shut the port down and then issue the **no shutdown** command in order to use the particular port again after a **security violation has occurred**.

```
S1(config)# interface FastEthernet 0/18
S1(config-if)# shutdown
Sep 20 06:57:28.532: %LINK-5-CHANGED: Interface FastEthernet0/18, changed state to
administratively down
S1(config-if)# no shutdown
Sep 20 06:57:48.186: %LINK-3-UPDOWN: Interface FastEthernet0/18, changed state to up
Sep 20 06:57:49.193: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/18, changed state to up
```




VLAN Trunking protocol (VTP)

- As networks **grow in size** and complexity, **centralized management of the VLAN** structure becomes crucial.
- VLAN Trunking Protocol (VTP) is a Layer 2 messaging protocol that provides a method for the distribution and management of the **VLAN database from a centralized server** in a network segment.
- **Any change** to the VLAN structure requires **further manual configuration**. One **incorrectly keyed** number causes inconsistencies in connectivity throughout the entire network.
- To resolve this issue, Cisco created **VTP** to **automate** many of the VLAN configuration functions.
- VTP ensures that VLAN configuration is **consistently maintained** across the network and reduces the task of **VLAN management and monitoring**.



- VTP is a **client/server messaging protocol** that **adds, deletes, and renames VLANs** in a single VTP domain.
- All switches under a **common administration** are **part of a domain**.
- Each domain has a **unique name**. VTP switches only **share VTP messages** with other switches in the **same domain**.
- Two different versions of VTP exist: **Version 1** and **Version 2**.
- All switches must be configured with the **same version**.
- VTP has three modes: **server, client, and transparent**.
- By default, all switches are **servers**. It is a good practice to have at **least two** switches configured **as servers** on a network, to provide **backup** and **redundancy**.



- ❑ **Server mode**—In VTP server mode, you can **create, modify, and delete** VLANs and specify other configuration parameters, such as **VTP version** and VTP running, for the entire VTP domain.

- ❑ **Client mode** —VTP clients behave the same way as VTP servers, but you **cannot create, change, or delete** VLANs on a VTP client.

- ❑ **Transparent**—VTP transparent switches do not participate in VTP.
 - A VTP transparent switch **does not advertise its VLAN** configuration and does not synchronize its VLAN configuration based on received advertisements, but transparent switches do **forward VTP advertisements** that they receive out their trunk ports in VTP Version 2.

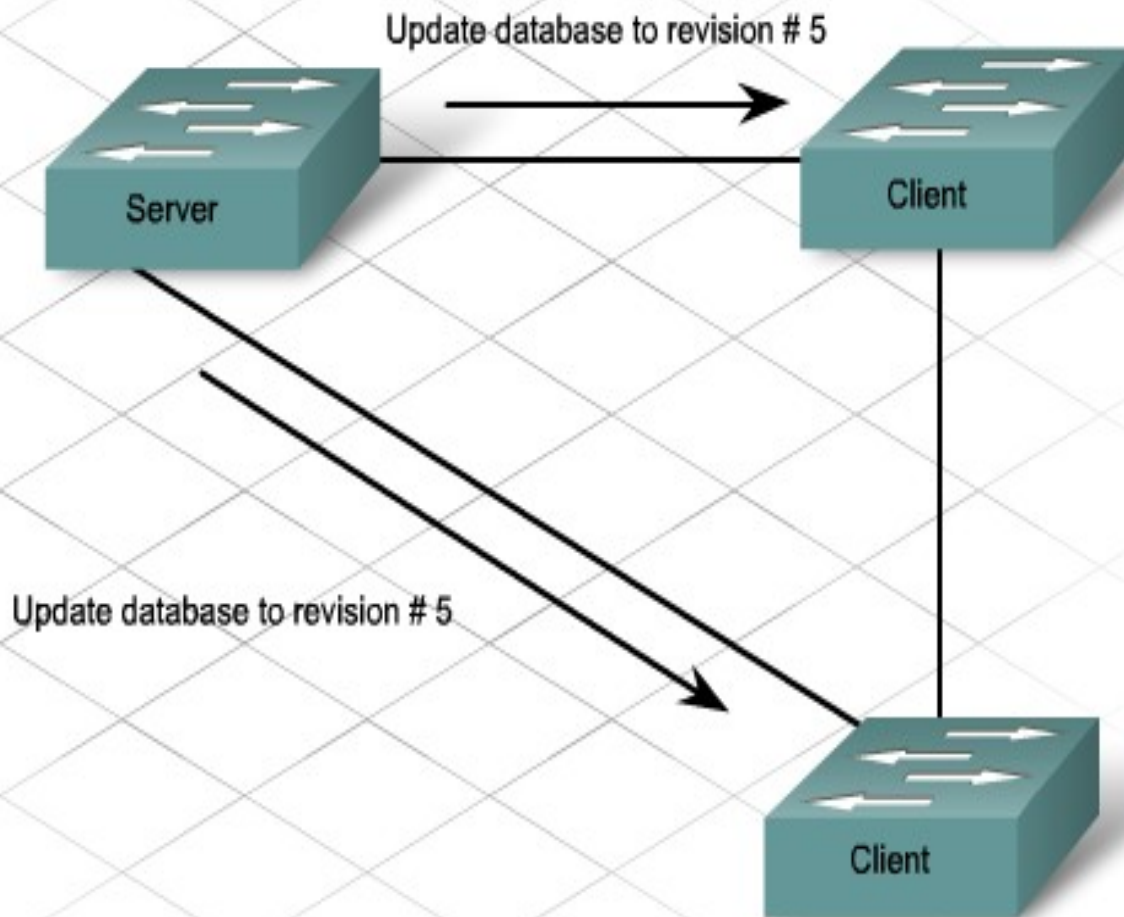


VTP

- With VTP, **each switch** advertises messages on its **trunk ports**.
- Messages include the **management domain**, **configuration revision number**, **known VLANs**, and **parameters for each VLAN**.
- These **advertisement frames** are sent to a **multicast** address so that all **neighbor devices** receive the frames.
- Each VTP switch **saves** a VLAN database in **NVRAM(non volatile random access memory)** that contains a revision number.
- If a VTP receives an update message that has a **higher revision number** than the one stored in the database, the switch updates its VLAN database with this new information.
- To protect some one to **insert incorrect** VLAN update, is to configure a **VTP password** to validate the switch.



VTP database





- VTP messages come in three varieties:
 - ✓ Summary advertisements
 - ✓ Subset advertisements
 - ✓ Advertisement requests.



Summary Advertisements

- Switches issue summary advertisements **every 5 minutes** or whenever a **change** to the **VLAN database occurs**.
- Summary advertisements contain the **current VTP domain name** and the **configuration revision number**.
- If VLANs are **added, deleted, or changed**, the **server increments** the **configuration revision number** and issues a summary advertisement.
- When a **switch receives** a summary advertisement packet, it compares the VTP domain name to its own VTP domain name.
- If the domain name is the **same**, the switch **compares** the **configuration revision number** to its own number. If it is **lower or equal**, the switch **ignores the packet**.
- If the revision number is **higher**, an advertisement request is sent.



❑ Subset Advertisements

- A subset advertisement **follows** the summary advertisement.
- A subset advertisement contains a **list of VLAN information**.
- The subset advertisement contains the **new VLAN information** based on the summary advertisement.
- If there are several VLANs, they require more than one subset advertisement.

❑ Advertisement Requests

- Switches use advertisement requests to **ask for VLAN information**.
- Advertisement requests are required if the **switch has been reset** or the **VTP domain name has been changed**.
- The switch receives a VTP summary advertisement with a higher configuration revision number than its own.



Configuring VTP

- If a switch in server mode issues an **update** with a higher revision number than the number currently in place, all **switches will modify** their databases to match the new switch.
- When **adding a new switch** to an **existing VTP domain**, use the following steps:

Step 1: Configure VTP off-line (version 1)

```
Switch(config)#vtp domain domain_name
Switch(config)#vtp mode {server | client | transparent}
Switch(config)#vtp password password
Switch(config)#end
Switch#copy running-config startup-config
```



Step 2: Verify the VTP configuration

```
Switch#sho vtp status

VTP Version                : 2
Configuration Revision      : 3
Maximum VLANs supported locally : 64
Number of existing VLANs    : 8
VTP Operating Mode          : Server
VTP Domain Name             : Cisco
VTP pruning Mode            : Disabled
VTP V2 Mode                  : Disabled
VTP Traps Generation        : Disabled
MDS digest : 0xF1 0xD3 0x88 0x0A 0xD2 0x09 0xA6 0xC5
Configuration last modified by 10.20.1.1 at 3-1-93 00:20:25
Local updater ID is 10.20.1.1 on interface V11 (first interface found)
```

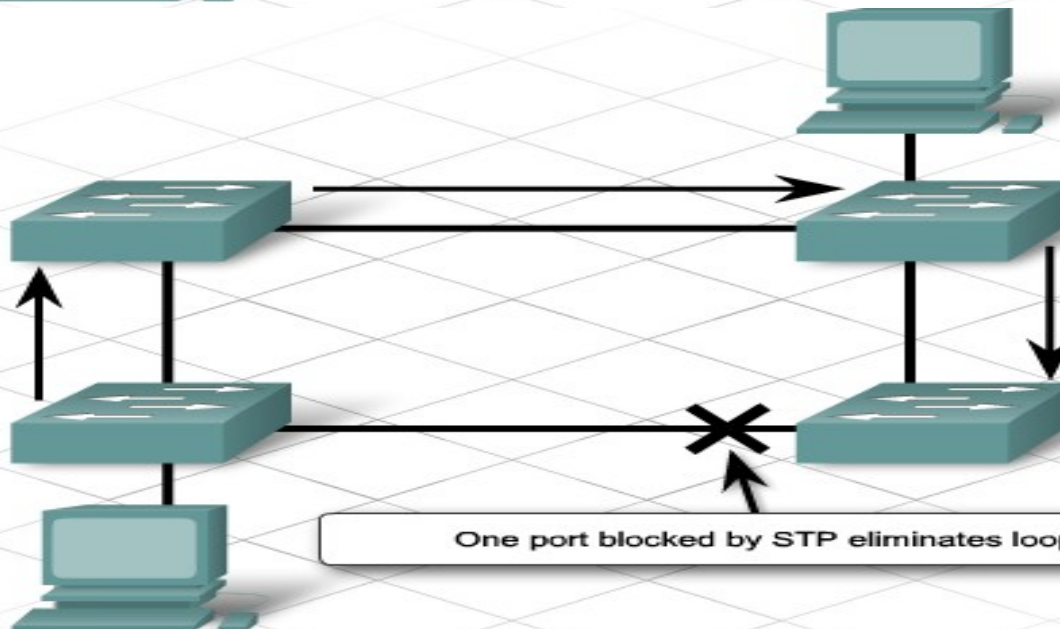
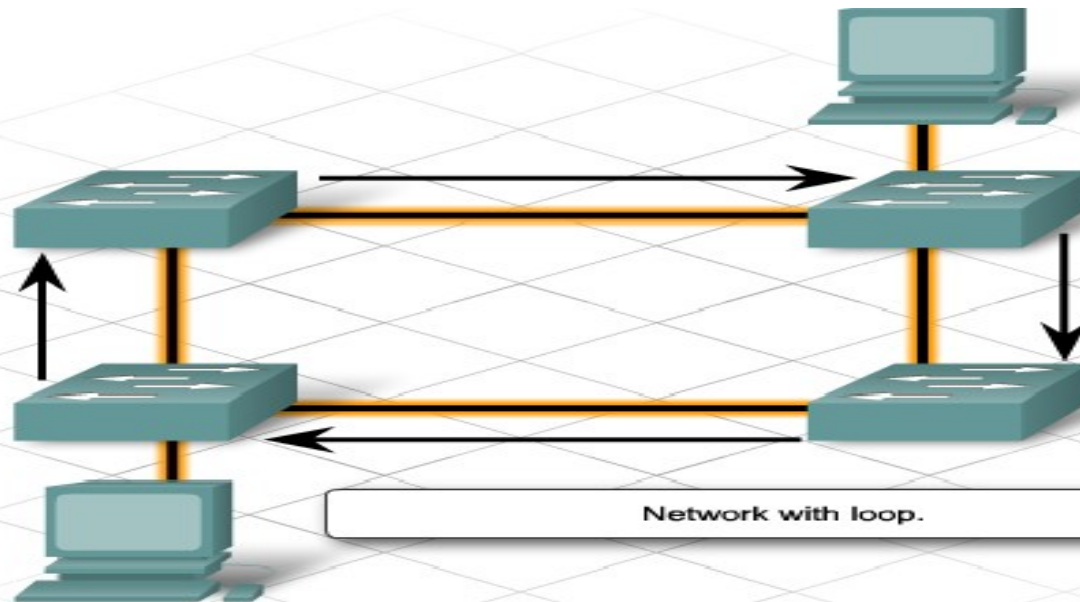
Step 3: Reboot the switch

```
Switch#reload
Switch#show vtp password
Switch#show vtp counters
```



Spanning Tree Protocol (STP)

- Spanning Tree Protocol (STP) provides a mechanism for disabling **redundant links** in a switched network.
- STP provides the redundancy required for reliability without **creating switching loops**.
- When switches are **first powered up** with STP enabled, they **check** the switched network for the **existence of loops**.
- Switches **detecting** a potential loop **block** some of the connecting ports, while leaving other ports active to forward frames.





- Switches are constantly checking the network **to ensure** that there are **no loops** and that all ports function as required.

To prevent switching loops, STP:

- ✓ **Forces** certain interfaces into a **standby** or **blocked state**
- ✓ Leaves other interfaces in a forwarding state
- ✓ Reconfigures the network by activating the appropriate **standby path**, if the forwarding path becomes unavailable



- In STP terminology, the term **bridge** is frequently used to refer to a **switch**.
- For example, the **Root Bridge** is the **primary switch** or focal point in the **STP** topology.
- The **root bridge** communicates with the other switches using **Bridge Protocol Data Units (BPDUs)**.
- BPDUs are frames that **multicast every 2 seconds** to all other switches.
- BPDUs contain information such as:
 - ✓ Identity of the source switch
 - ✓ Identity of the source port
 - ✓ Cumulative cost of path to root bridge
 - ✓ Value of aging timers
 - ✓ Value of the hello timer



- As a **switch powers on**, each port cycles through a series of **four states**:
 - Blocking,
 - Listening,
 - Learning, and
 - Forwarding.

- A fifth state, **disabled**, indicates that the administrator has **shut down the switch port**.



- When a **switch powers on**, it first goes into a **blocking state** to immediately **prevent** the formation of a **loop**.
- It then changes to **listening mode**, so that it receives **BPDUs** from neighbor switches.
- After processing this information the **switch determines** which **ports can forward** frames **without creating a loop**.
- If the **port** can forward frames, it changes to **learning mode**, and then to **forwarding mode**.
- **Access ports do not create loops** in a switched network and always transition to **forwarding** if they have a host attached.
- **Trunking ports potentially create a looped network** and transition to either a **forwarding or blocking state**.



- STP uses root switch, to **determine** which **ports to block** and which **ports to put into forwarding** state.
- The **root bridge** sends out BPDUs containing network topology information to **all other switches**. This information allows the network to reconfigure itself in the event of a failure.
- There is **only one root bridge** on each network, and it is elected based on the **bridge ID (BID)**.
- The **bridge priority** value plus the **MAC address** creates the BID.
- **Bridge priority has a default value of 32,768**. If a switch has a MAC address of AA-11-BB-22-CC-33, the BID for that switch would be: 32768: AA-11-BB-22-CC-33.



Root Bridge

- The root bridge is based on the **lowest BID** value. Since switches typically use the **same default priority value**, the switch with the **lowest MAC address** becomes the root bridge.
- As each **switch powers** on, it **assumes that it is the root bridge**, and sends out BPDUs containing its BID.
- For example, if S2 advertises a root ID that is a lower number than S1, S1 stops the advertisement of its root ID and accepts the root ID of S2. S2 is now the root bridge.
- STP designates three types of ports: **root ports, designated ports, and blocked ports.**



Root Port

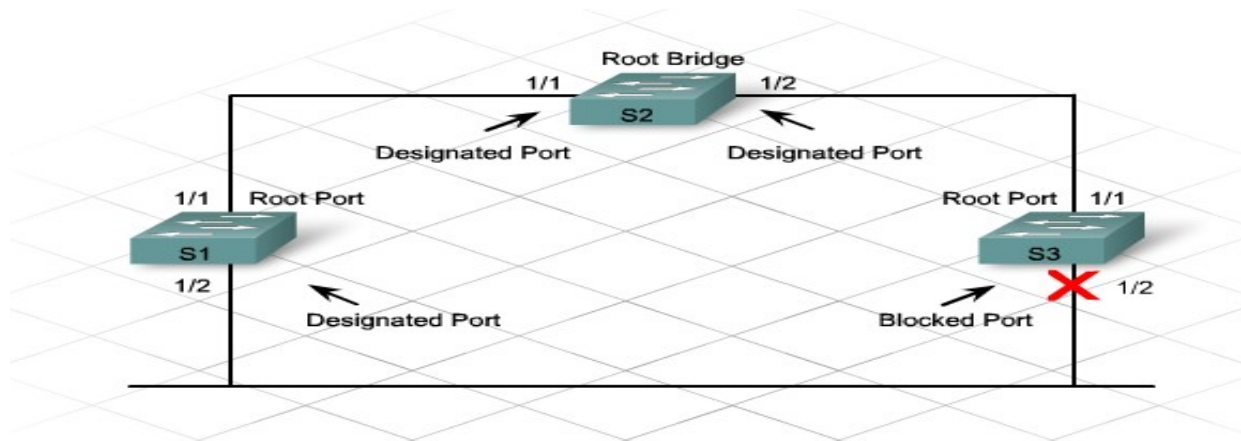
- The port that provides the **least cost path** back to the root bridge **becomes the root port**. Switches calculate the least cost path using the bandwidth cost of each link required to reach the root bridge.

Designated Port

- A designated port is a port that **forwards traffic toward the root bridge** but **does not connect to the least cost path**.

Blocked Port

- A blocked port is a port that does not forward traffic.

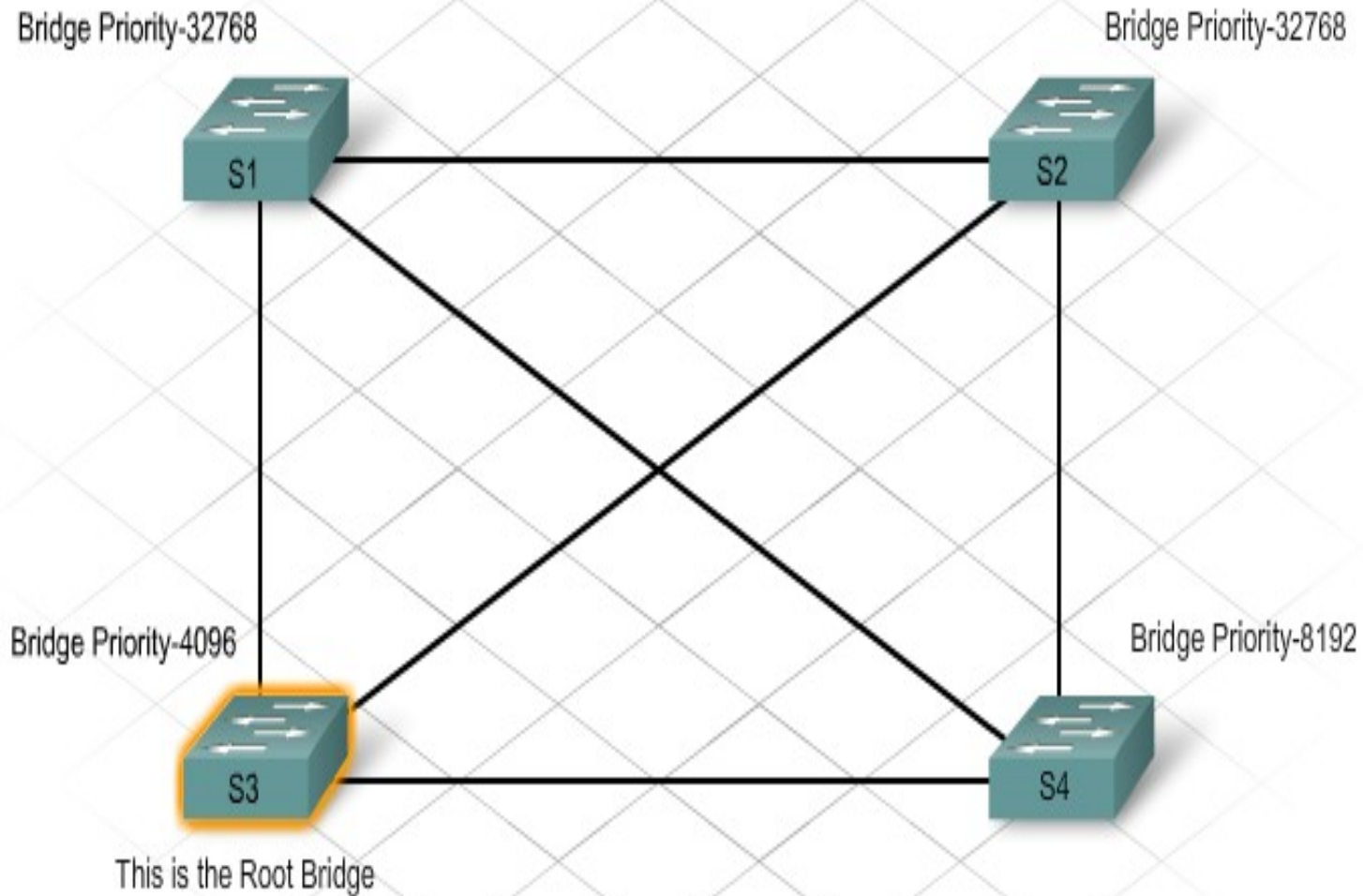




- Before configuring STP, the **network technician plans** and evaluates the network in order to select the **best switch to become the root of the spanning tree.**
- If the root switch goes to the default lowest MAC address, forwarding might not be optimal.
- To specify the root bridge, the BID of the chosen switch is **configured with the lowest priority value.**
- The bridge priority command is used to configure the bridge priority.



- The **range for the priority** is from 0 to 65535, the default value is 32768.
- To set priority:
- S3(config)#spanning-tree vlan 1 priority 4096
- To restore priority to default:
- S3(config)#no spanning-tree vlan 1 priority





Spanning tree in Hierarchical Network

- After **establishing the root bridge, root ports, designated ports, and blocked ports**, STP sends BPDUs throughout the switched network at 2-second intervals.
- **STP continues** to listen to these BPDUs to ensure that no links **fail** and **no new loops** appear.
- If a link failure occurs, STP recalculates by:
 - Changing some blocked ports to forwarding ports
 - Changing some forwarding ports to blocked ports
 - Forming a new STP tree to maintain the loop-free integrity of the network



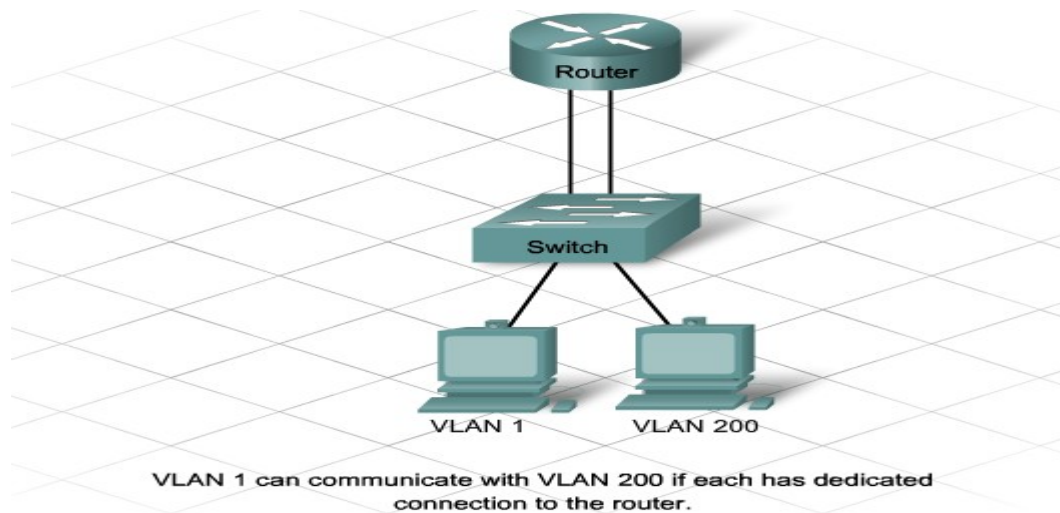
- When a link goes down, **STP detects the failure** and recalculates the best paths across the network.
- This calculation and transition period takes about 30 to 50 seconds on each switch. **During this recalculation**, no user **data passes** through the recalculating ports.
- Some user **applications time out** during the **recalculation period**, which can result in lost productivity and revenue.
- Frequent STP recalculations negatively impact uptime.



- There are **several useful commands** used to verify spanning tree operation.
- **show spanning-tree** - Displays bridge ID, and port states
- **show spanning-tree summary** - Displays a summary of port states
- **show spanning-tree root** - Displays the status and configuration of the root bridge
- **show spanning-tree detail** - Displays detailed port information
- **show spanning-tree interface** - Displays STP interface status and configuration
- **show spanning-tree blockedports** - Displays blocked ports

Inter-VLAN Routing

- Although VLANs extend to span multiple switches, only members of the same VLAN can communicate.
- A **Layer 3 device** provides connectivity between **different VLANs**.
- This arrangement enables the network administrator to strictly control the type of traffic that flows from one VLAN to another.
- One method of accomplishing the inter-VLAN routing requires a separate interface connection to the Layer 3 device for each VLAN.

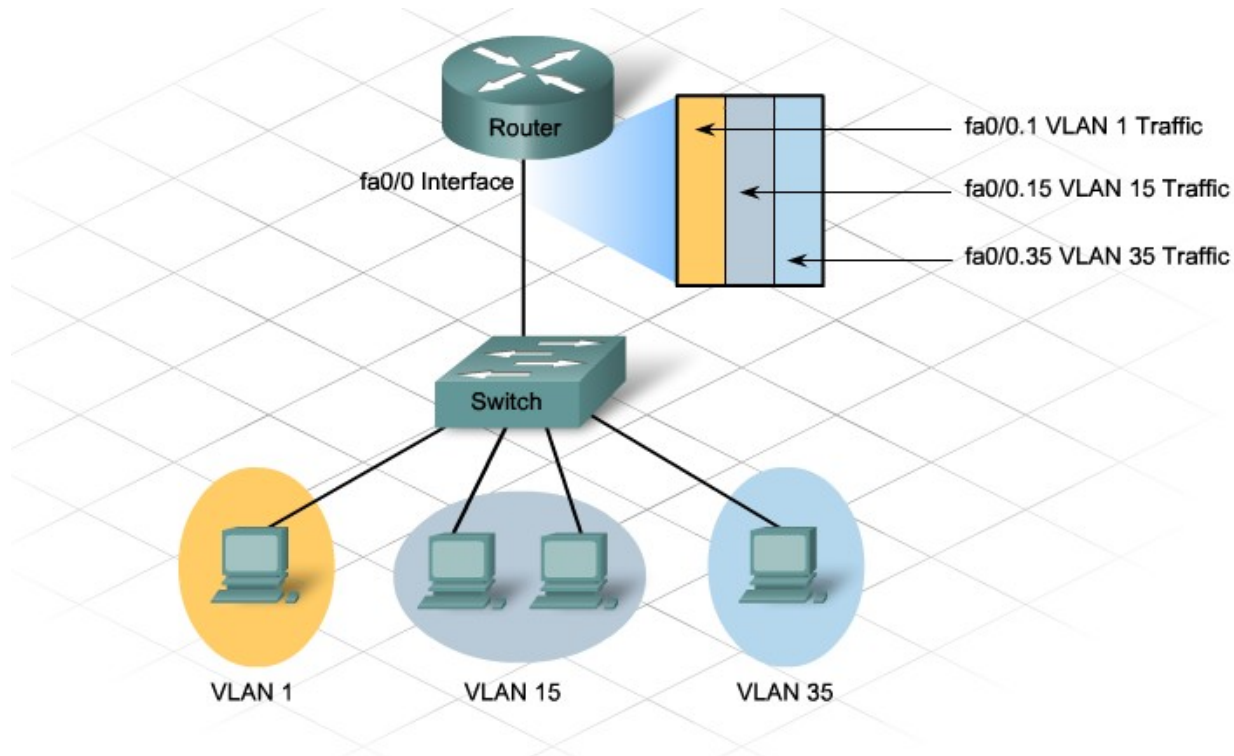




- Another method for providing connectivity between different VLANs requires a feature called **subinterfaces**.
- Subinterfaces logically **divide one physical interface** into multiple logical pathways.
- Configure one pathway or subinterface for each VLAN.
- To support **inter-VLAN communication** using subinterfaces requires configuration on **both the switch and the router**.



- **Switch**
- Configure the switch interface as an 802.1Q trunk link.
- **Router**
- Select a router interface with a minimum of a 100Mbps FastEthernet
- Configure subinterfaces that support 802.1Q encapsulation.
- Configure one subinterface for each VLAN.
- A subinterface allows each VLAN to have its own **logical pathway** and default gateway into the router.



- The host from the sending VLAN forwards traffic to the router using the default gateway.
- The subinterface for the VLAN specifies the default gateway for all hosts in that VLAN.
- The router locates the destination IP address and does a routing table lookup.



- If the **destination VLAN is on the same** switch as the source VLAN, the router forwards the traffic back down to the source switch using the subinterface parameters of the destination VLAN ID. This type of configuration is often referred to as a **router-on-a-stick**.
- If the exit interface of the router is 802.1Q-compatible, the frame retains its 4-byte VLAN tag.
- If the outbound interface is not 802.1Q-compatible, the router strips the tag from the frame and returns the frame to its original Ethernet format.



To configure inter-VLAN routing, use the following steps:

1. Configure a trunk port on the switch.

```
Switch(config)#interface fa0/2
```

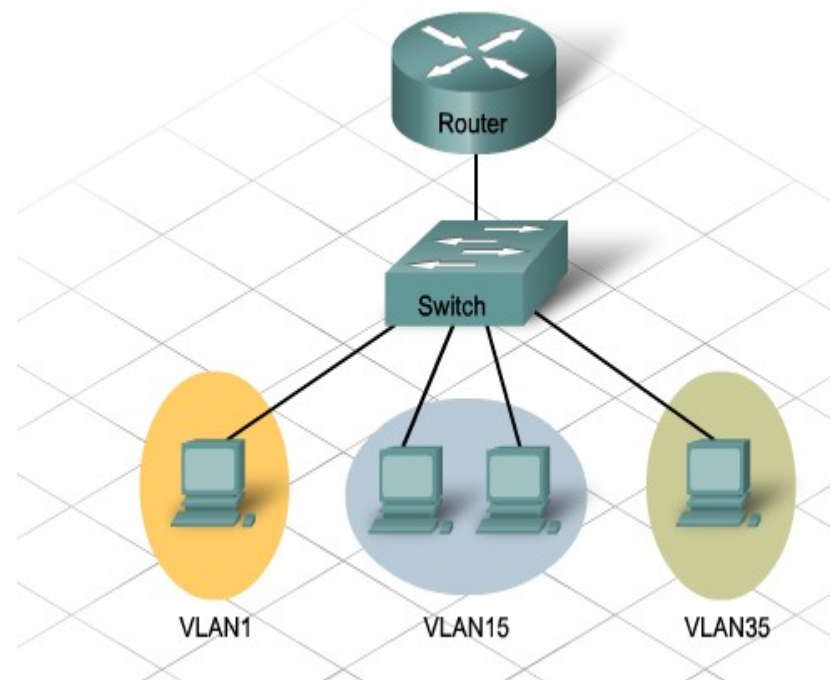
```
Switch(config-if)#switchport mode trunk
```

2. On the router, configure a FastEthernet interface with no IP address or subnet mask.

```
Router(config)#interface fa0/1
```

```
Router(config-if)#no ip address
```

```
Router(config-if)#no shutdown
```





- 3. On the router, configure one subinterface with an IP address and subnet mask for each VLAN. Each subinterface has an 802.1Q encapsulation.
- Router(config)#interface fa0/0.10
- Router(config-subif)#encapsulation dot1q 10
- Router(config-subif)#ip address 192.168.10.1 255.255.255.0
- 4. Use the following commands to verify the inter-VLAN routing configuration and functionality.
- Switch#show trunk
- Router#show ip interfaces
- Router#show ip interfaces brief
- Router#show ip route



END